UNIVERSITY OF MALAYA

EXAMINATION FOR THE DEGREE OF MASTER OF DATA SCIENCE

ACADEMIC SESSION 2018/2019 : SEMESTER II

WQD7006 : Machine Learning for Data Science

June 2019 Time : 3 hours

INSTRUCTIONS TO CANDIDATES :

Answer **ALL** questions (50 marks).

(This question paper consists of 5 questions on 6 printed pages)

1. (a) Differentiate the following terms/concepts in a **TABULAR** format.

   i. Classification and Prediction
   ii. Signal and noise
   iii. Generative and Discriminative models
   iv. Bias and variance
   v. Local and global minima

   (5 marks)

   (b) Ali uses either car, bus or MRT to work. If he goes by car, 50% chance he will be late, if he goes by bus then 20% he will be late and if it's MRT, only 1% he will be late.

   i. You are given that Ali is late one day. His boss wants to estimate the probability that Ali came by car. He gave the prior probability for each mode of transportation 1/3. What is Ali's boss probability (or estimate) that Ali came by car?

   (4 marks)

   ii. Now suppose Ali's colleague knows that Ali usually takes the MRT, never takes the bus and 10% of the time takes the car. What is the colleague's probability that Ali drove to work that day, given he was late?

   (1 mark)

2. (a) You run gradient descent for 15 iterations with $\alpha = 0.3$ and compute $J(\theta)$ after each iteration. You find that value of $J(\theta)$ increases over time. Comment on the choice of alpha in this case, if it is appropriate, should be larger or smaller, giving an explanation. (1 mark)

   (b) Consider the problem of predicting how well a student does in his/ her second year of college university, given how well they did in their first year. Specifically, let x be the number of "A" grades that a student receives in their first year. We would like to predict the value of y which we define as the number of "A" grades that they get in their second year.

   Refer to the following dataset, each row is one training example. Our hypothesis (in linear regression) is $h_\theta(x) = \theta_0 + \theta_1 x$, and we use m to denote the training examples.

x	y
3	4
2	1
4	3
0	1

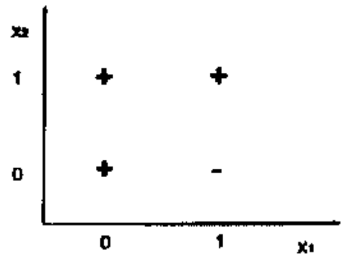
i. What is the value of m?

ii. Our definition of the cost function is $J(\theta_0,\theta_1)=\frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^{(i)})-y^{(i)})^2$
What is $J(0,1)$?

iii. Suppose we set $\theta_0=0,\theta_1=1.5$. What is $h_\theta(2)$?

(4 marks)

(c) We are interested in predicting whether a person makes over 50K a year, and we model the two features with two boolean variables X1, X2 ∈ {0,1}, and label Y ∈ {0,1} where Y = 1 indicates a person makes over 50K. Figure below shows three positive samples ("+" for Y = 1) and one negative sample ("-" for Y = 0). Answer the following questions:



i. For the above scenario, which model would be better in predicting: linear or logistic regression? Why? (2 marks)

ii. Is there any logistic regression classifier using X1 and X2 that can perfectly classify the examples in the figure above? Explain. (2 marks)

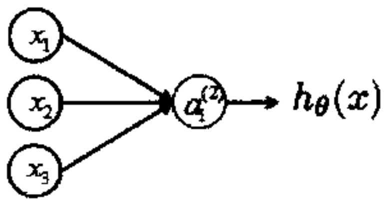
iii. If we change the label of point (0,1) from "+" to "-", will there be a perfect logistic regression classifier? (1 mark)

3. The dataset below is used to learn a decision tree which predicts if people pass machine learning (Yes or No), based on their previous GPA (High, Medium, or Low) and whether or not they studied.

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

i. What is the entropy H(Passed)? (2 marks)
ii. What is the entropy H(Passed | GPA)? (3 marks)
iii. What is the entropy H(Passed | Studied)? (3 marks)
iv. Draw the full decision tree that would be learned for this dataset. (2 marks)

4. Assume a node with no bias, as below:



a. The node has three inputs x = (x1, x2, x3) that receive only binary signals (either 0 or 1). Determine how many different input patterns this node can receive?

(1 mark)

b. Suppose that the weights corresponding to the three inputs have the following values:
$w_1 = 2$, $w_2 = -4$, $w_3 = 2$

and the activation of the unit is given by the step-function:

$$\varphi(a_i^{(j)}) = \begin{cases} 1 & \text{if } a_i^{(j)} \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

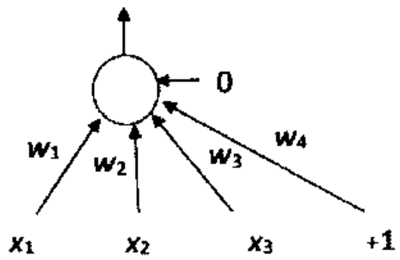
Note: $a_i^{(j)}$ = "Activation" of unit *i* in layer *j*

Calculate what will be the output value y of the unit for each of the following input patterns:

Pattern, P_i	P_1	P_2	P_3
[redacted]			
X_2	*1*	*1*	*0*
[redacted]			

(4 marks)

c. Consider a Perceptron with 3 inputs, x1, x2, x3, and one output unit that uses a linear threshold unit (LTU) as its activation function. The initial weights are w1 = 0.2, w2 = 0.7, w3 = 0.9, and bias w4 = – 0.7 (the LTU uses a fixed threshold value of 0). Hence we have:



Given the inputs x1 = 0, x2 = 0, x3 = 1 and the above weights, this Perceptron outputs 1. What are the four (4) updated weights' values after applying the Perceptron Learning Rule with this input, actual output = 1, target output = 0, and learning rate = 0.2?

(5 marks)

5. Answer the following short questions: (Each question carries 1 mark, unless specified)

(a) What is supervised learning?

(b) Name one classifier for Probabilistic and one for Clustering.

(c) What is reinforcement learning?

(d) What may happen if the step-size for a Gradient Descent algorithm is too big?

(e) What are the TWO approaches to avoid overfitting in decision trees?

(f) Draw a multi-layer perceptron.

(g) What does a 10-fold cross validation mean?

(h) Name one field/area where clustering would be useful?

(i) Name one popular technique to measure distance in K-means clustering.

(j) In one sentence, explain mechanism behind bagging.

(10 marks)

END